Carleton University

Course Project: SCARA Robot Report

MECH 4503A: An Introduction to Robotics

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1 Introduction

The Selective Compliance Assembly Robot Arm (SCARA) is a widely used industrial manipulator first proposed by Hiroshi Makino at Yamanashi University, Japan, in 1978 and subsequently commercialized in the early 1980s. Its defining characteristic is *selective compliance*: it is rigid in the vertical (Z) direction yet compliant in the horizontal (XY) plane. This property makes the SCARA uniquely suited to pick-and-place operations, printed circuit board assembly, packaging, and precision insertion tasks, where lateral flexibility accommodates minor misalignments while vertical rigidity resists assembly forces.

A standard SCARA robot possesses four degrees of freedom (DOF): two revolute joints in the horizontal plane (joints 1 and 2), one prismatic joint for vertical translation (joint 3), and one revolute joint for end-effector rotation (joint 4), as shown in Figure 1.

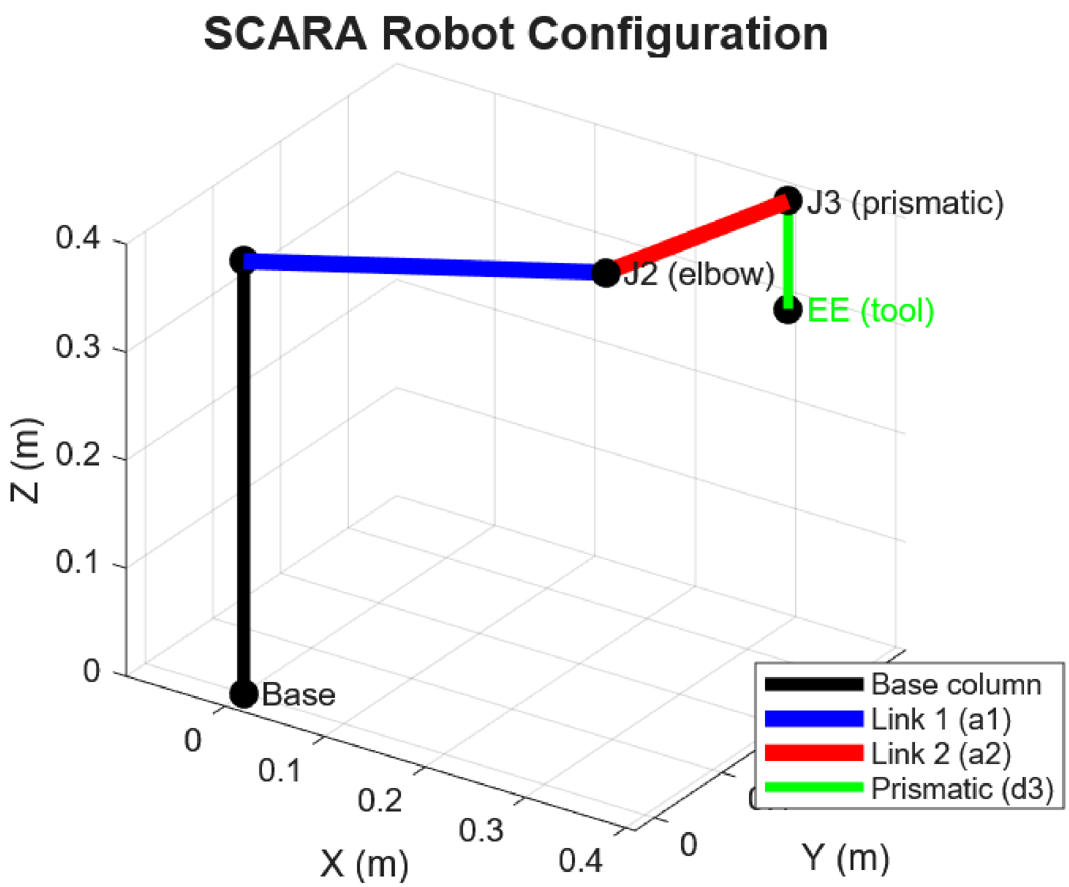

Figure 1: Typical 4-DOF SCARA robot configuration

This report derives the complete kinematic and dynamic models for a 4-DOF SCARA, characterises its work envelope, and provides MATLAB code for visualization. The derivations follow the Denavit-Hartenberg (DH) convention and the Lagrangian mechanics formulation.

The report is structured as follows: Section 2 describes the mechanical architecture; Section 3 derives forward kinematics; Section 4 derives inverse kinematics; Section 5 analyses velocity kinematics via the Jacobian; Section 6 characterises the work envelope; Section 7 derives the full equations of motion; and Appendix A provides MATLAB visualization code.

2 SCARA Robot Architecture

2.1 Mechanical Configuration

The canonical 4-DOF SCARA robot consists of a fixed base, two horizontal revolute links of lengths a_1 and a_2, a vertical prismatic joint, and a final revolute wrist joint. Table 1 summarises the joint types and their physical roles.

The relative pose between consecutive coordinate frames is defined by four elementary transformations. The basic rotation and translation matrices used to construct the link transformations are defined as:

$$\mathrm{Rot}_z(\theta_i) = \begin{bmatrix} cos\theta_i & -sin\theta_i & 0 & 0 \\ sin\theta_i & cos\theta_i & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathrm{Trans}_z(d_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\mathrm{Trans}_x(a_i) = \begin{bmatrix} 1 & 0 & 0 & a_i \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad \mathrm{Rot}_x(\alpha_i) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & cos\alpha_i & -sin\alpha_i & 0 \\ 0 & sin\alpha_i & cos\alpha_i & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

These matrices represent the four physical parameters that describe the robot's links and joints:

- θ_i **(Joint angle):** The angle between the x_{i-1} and x_i axes, measured around the z_i axis.
- d_i **(Link offset):** The distance along the z_i axis between the x_{i-1} and x_i axes.
- a_i **(Link length):** The distance along the x_i axis between the z_{i-1} and z_i axes.
- α_i **(Link twist):** The angle between the z_{i-1} and z_i axes, measured around the x_i axis.

Joint	Type	Symbol	Motion	Role
1	Revolute	θ_1	Rotation about Z_0	Shoulder rotation in XY plane
2	Revolute	θ_2	Rotation about Z_1	Elbow rotation in XY plane
3	Prismatic	d_3	Translation along Z_2	Vertical (Z) positioning
4	Revolute	θ_4	Rotation about Z_3	End-effector spin / tool orientation

Table 1: SCARA joint summary.

2.2 Joint Parameters and DH Convention

The Denavit-Hartenberg (DH) parameterisation assigns four scalar parameters to each link-joint pair: link length a_i, link twist α_i, joint offset d_i, and joint angle θ_i. The parameters for this SCARA model are summarized in Table 2, and the corresponding coordinate frame assignments are illustrated in Figure 2.

Joint i	a_i (m)	α_i (rad)	d_i (m)	θ_i (rad)
1	a_1	0	d_1 (const.)	θ_1 (var.)
2	a_2	π	0	θ_2 (var.)
3	0	0	d_3 (var.)	0
4	0	0	d_4 (const.)	θ_4 (var.)

Table 2: DH parameters for the standard 4-DOF SCARA robot.

Typical industrial values for this model are $a_1 = 0.300$ m, $a_2 = 0.200$ m, and a prismatic stroke for joint 3 of approximately 0.150 m.

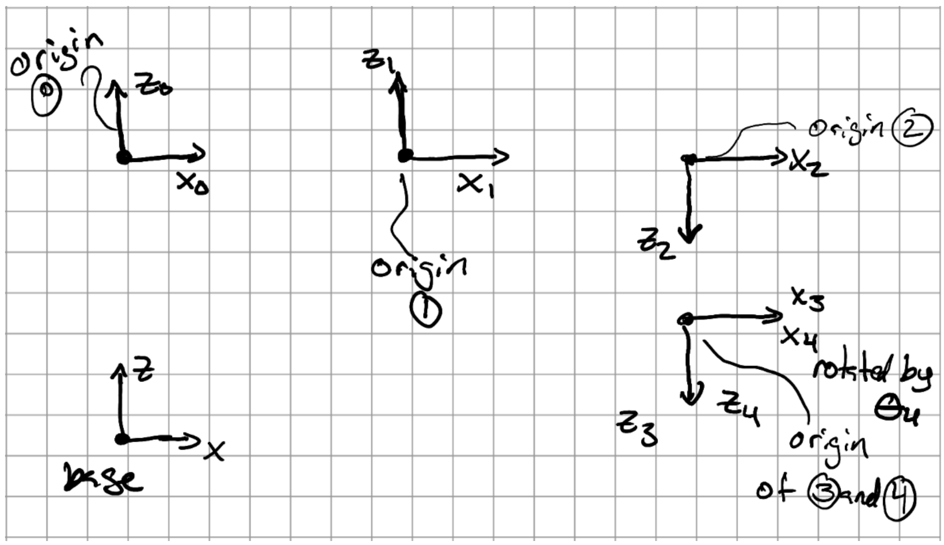

Figure 2: Detailed coordinate frame assignments. Note the 180° twist ($\alpha_2 = \pi$) at Joint 2, which directs the Z_2, Z_3, and Z_4 axes downward.

The frame assignments follow the standard DH convention with one critical specification: frame 2 has its Z_2 axis *anti-parallel* to Z_1. As shown in Figure 2, the link twist $\alpha_2 = \pi$ signifies a 180° rotation of the Z-axis about X_1 when transitioning from frame 1 to frame 2.

As a result, Z_2 points downward, ensuring that:

1. Joint 3 (prismatic) translates in the downward direction ($-Z_0$) as the variable d_3 increases.
2. The (3,3) entry of the T_1^2 transformation matrix is -1.

This anti-parallel Z convention is the defining feature of this SCARA model and serves as the source of the negative signs appearing throughout the kinematic and dynamic derivations in Sections 3 and 7.

3 Forward Kinematics

3.1 Denavit-Hartenberg Transformation Matrix

Using the standard DH convention, the homogeneous transformation from frame $i-1$ to frame i is as shown below.

$$T_{i-1}^{i} = \mathrm{Rot}_z(\theta_i)\,\mathrm{Trans}_z(d_i)\,\mathrm{Trans}_x(a_i)\,\mathrm{Rot}_x(\alpha_i) \tag{3.1a}$$

Expanding this product yields the general 4×4 transformation:

$$T_{i-1}^{i} = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_i & \sin\theta_i\sin\alpha_i & a_i\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_i & -\cos\theta_i\sin\alpha_i & a_i\sin\theta_i \\ 0 & \sin\alpha_i & \cos\alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.1b}$$

3.2 Individual Link Transformations

Substituting the DH parameters from Table 2 into Equation (3.1b) gives the following individual transformations. Let $c_i = \cos\theta_i$ and $s_i = \sin\theta_i$ for compactness.

Link 1 (revolute, $\alpha_1 = 0$):

$$T_0^1 = \begin{bmatrix} c_1 & -s_1 & 0 & a_1c_1 \\ s_1 & c_1 & 0 & a_1s_1 \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.2}$$

Link 2 (revolute, $\alpha_2 = \pi$):

$$T_1^2 = \begin{bmatrix} c_2 & s_2 & 0 & a_2c_2 \\ s_2 & -c_2 & 0 & a_2s_2 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.3}$$

Link 3 (prismatic, $\theta_3 = 0, a_3 = 0, \alpha_3 = 0$):

$$T_2^3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.4}$$

Link 4 (revolute, $a_4 = 0, \alpha_4 = 0$):

$$T_3^4 = \begin{bmatrix} c_4 & -s_4 & 0 & 0 \\ s_4 & c_4 & 0 & 0 \\ 0 & 0 & 1 & d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.5}$$

Note: although $\alpha_4 = 0$ gives $+1$ at position (3,3) in T_3^4, the end-effector Z-axis still points *downward* in the base frame because the Z-flip introduced by $\alpha_2 = \pi$ propagates through the entire kinematic chain. The tool therefore faces downward by default, which is the correct physical configuration for a SCARA performing vertical insertion tasks.

3.3 Full End-Effector Transformation

The overall transformation from the base frame (frame 0) to the end-effector (frame 4) is the sequential product:

$$T_0^4 = T_0^1 T_1^2 T_2^3 T_3^4 \tag{3.6}$$

Carrying out the multiplication (using $c_{12} = \cos(\theta_1 + \theta_2)$ and $s_{12} = \sin(\theta_1 + \theta_2)$):

$$T_0^4 = \begin{bmatrix} \cos(\theta_1+\theta_2-\theta_4) & \sin(\theta_1+\theta_2-\theta_4) & 0 & a_1c_1 + a_2c_{12} \\ \sin(\theta_1+\theta_2-\theta_4) & -\cos(\theta_1+\theta_2-\theta_4) & 0 & a_1s_1 + a_2s_{12} \\ 0 & 0 & -1 & d_1 - d_3 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3.7}$$

The end-effector position in the base frame is therefore:

$$\begin{aligned} p_x &= a_1 \cos\theta_1 + a_2 \cos(\theta_1 + \theta_2) \\ p_y &= a_1 \sin\theta_1 + a_2 \sin(\theta_1 + \theta_2) \\ p_z &= d_1 - d_3 - d_4 \end{aligned} \tag{3.8}$$

The end-effector orientation about Z is $\phi = \theta_1 + \theta_2 - \theta_4$ (the subtraction appears because $\alpha_2 = \pi$ reverses the sense of joint 4 relative to the tool). This is a key feature of the SCARA robot, where position and orientation are partially decoupled, which simplifies both planning and inverse kinematics.

The validity of the forward kinematic model in Equation 3.8 is demonstrated through a numerical simulation of a sinusoidal joint trajectory. Figure 3 illustrates the resulting continuous path of the end-effector in 3D space.

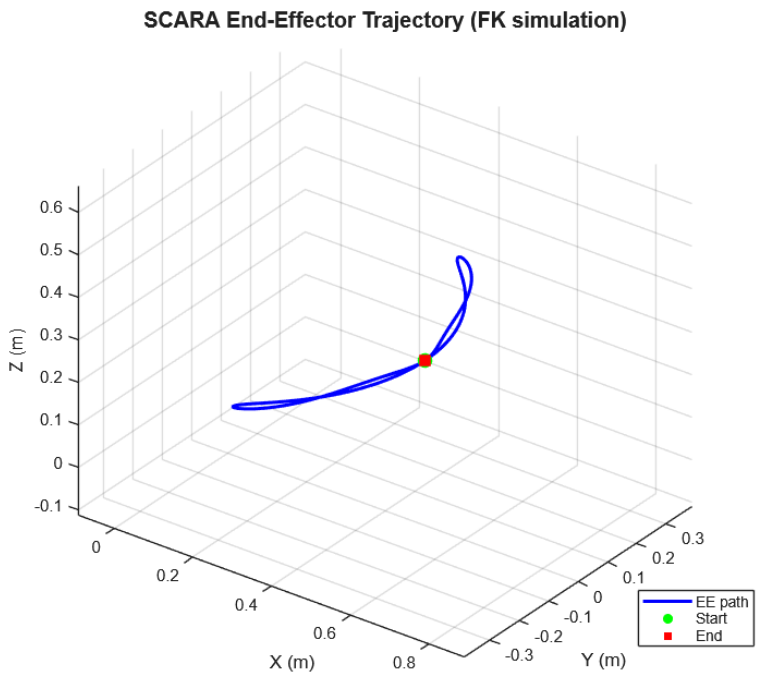

Figure 3: SCARA End-Effector Trajectory generated via Forward Kinematics (Eq. 3.8) for sinusoidal joint inputs.

4 Inverse Kinematics

4.1 Geometric Approach

Given a desired end-effector position (p_x, p_y, p_z) and tool orientation angle ϕ, we seek the joint vector $\mathbf{q} = [\theta_1, \theta_2, d_3, \theta_4]^T$. The derivation follows the geometric method.

Step 1 — Vertical joint (d_3): The prismatic joint directly sets the height. Based on the frame assignments where the Z-axis flips at joint 2:

$$d_3 = d_1 - p_z - d_4 \tag{4.1}$$

Step 2 — Elbow angle (θ_2): Squaring and summing the x and y position components:

$$p_x^2 + p_y^2 = a_1^2 + a_2^2 + 2a_1a_2\cos(\theta_2) \tag{4.2a}$$

Solving for $\cos(\theta_2)$ via the Law of Cosines:

$$\cos(\theta_2) = \frac{p_x^2 + p_y^2 - a_1^2 - a_2^2}{2a_1a_2} \tag{4.2b}$$

Define $D = \cos(\theta_2)$. For a reachable point, we require $|D| \leq 1$. The joint angle is then:

$$\theta_2 = \text{atan2}\left(\pm\sqrt{1 - D^2}, D\right) \tag{4.2c}$$

The $\pm$ sign provides the two solution branches (elbow-up and elbow-down).

Step 3 — Shoulder angle (θ_1): Using the two-argument arctangent to find the shoulder orientation relative to the target:

$$\theta_1 = \text{atan2}(p_y, p_x) - \text{atan2}(a_2\sin\theta_2, a_1 + a_2\cos\theta_2) \tag{4.3}$$

Step 4 — Wrist angle (θ_4): From the orientation constraint $\phi = \theta_1 + \theta_2 - \theta_4$:

$$\theta_4 = \theta_1 + \theta_2 - \phi \tag{4.4}$$

4.2 Elbow-Up and Elbow-Down Solutions

The sign choice in Equation 4.2c yields two geometrically distinct configurations:

- **Elbow-up (positive sign):** $\theta_2 > 0$; the elbow is "above" the line joining the shoulder to the wrist. This is often preferred in assembly tasks to avoid table collisions.
- **Elbow-down (negative sign):** $\theta_2 < 0$; the elbow is "below" the line. This configuration is used when the workspace requires a different approach direction.

Both solutions are physically valid when $|D| < 1$. Practitioners select the branch based on obstacle constraints or proximity to a preferred configuration. When $D = \pm 1$ the robot is at a boundary singularity (fully extended or fully folded) and the solution is unique.

If $|D| > 1$, the desired position lies outside the reachable workspace and no real solution exists; the trajectory planner must reject the target or re-plan to a reachable point before invoking IK.

Regarding sign convention: θ_2 is measured in the same rotational sense as θ_1 (right-hand rule about each joint's downward-pointing Z-axis per the DH convention). Elbow-up ($\theta_2 > 0$) means the second link bends in the same angular direction as the first; elbow-down ($\theta_2 < 0$) means it bends in the opposite direction. The labels "up" and "down" describe the elbow position relative to the straight line joining the shoulder axis to the wrist point, as viewed from above.

5 Jacobian and Velocity Kinematics

5.1 Jacobian Matrix Derivation

The Jacobian J maps joint velocities $\dot{\mathbf{q}}$ to the end-effector velocity twist $[\dot{x}, \dot{y}, \dot{z}, \omega_x, \omega_y, \omega_z]^T$. For a SCARA robot, the angular velocity possesses a component only about the Z-axis, and the prismatic joint contributes exclusively to linear velocity along Z:

$$\dot{\xi} = J(\mathbf{q})\dot{\mathbf{q}} \tag{5.1}$$

Computing column-by-column using the geometric formulas for revolute joints ($\mathbf{J}_{v_i} = \mathbf{z}_{i-1} \times (\mathbf{p}_e - \mathbf{p}_{i-1})$) and prismatic joints ($\mathbf{J}_{v_i} = \mathbf{z}_{i-1}$):

$$J = \begin{bmatrix} -(a_1 s_1 + a_2 s_{12}) & -a_2 s_{12} & 0 & 0 \\ (a_1 c_1 + a_2 c_{12}) & a_2 c_{12} & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -1 \end{bmatrix} \tag{5.2}$$

Equation 5.2 represents the 6×4 geometric Jacobian. Rows 1–3 correspond to linear velocity components, while rows 4–6 represent angular velocity. The -1 entry in $J(3,3)$ remains consistent with the downward prismatic joint convention established in the DH parameters.

5.2 Singularity Analysis

Singularities occur when the Jacobian loses rank, indicating the manipulator has lost a degree of freedom in Cartesian space. For the SCARA, the determinant of the 2×2 planar velocity sub-matrix is:

$$\det(J_{xy}) = a_1 a_2 \sin(\theta_2) \tag{5.3}$$

A singularity occurs when $\sin(\theta_2) = 0$, specifically at $\theta_2 = 0$ or $\theta_2 = \pi$. These represent the fully extended and fully folded configurations, respectively, where radial motion is impossible. Avoiding these configurations is critical for trajectory planning to maintain full manipulability.

6 Work Envelope

6.1 Reachable and Dexterous Workspace

The reachable workspace consists of all Cartesian positions achievable by at least one joint configuration, while the dexterous workspace is the subset reachable with arbitrary end-effector orientation. For the SCARA, the horizontal workspace is an annular sector. The radial distance r satisfies:

$$|a_1 - a_2| \leq r \leq a_1 + a_2 \tag{6.1}$$

where $r_{min} = |a_1 - a_2|$ and $r_{max} = a_1 + a_2$. The vertical range is defined by the prismatic stroke:

$$d_1 - d_{3,max} - d_4 \leq z \leq d_1 - d_{3,min} - d_4 \tag{6.2}$$

The 3D reachable volume is a cylindrical annulus (a torus-like section) swept by the horizontal 2D ring over the vertical stroke. The total volume V is approximated by:

$$V = \pi(r_{max}^2 - r_{min}^2)(z_{max} - z_{min}) \tag{6.3}$$

6.2 Workspace Parameters

Table 3 provides representative parameters for a typical industrial SCARA robot.

The visualization of the 3D annular cylindrical workspace is produced by a MATLAB script provided in Appendix A. Figure 3 shows the reachable workspace boundary viewed from above (XY projection) and in perspective (3D annular cylinder), confirming the analytical limits derived in Equations 6.1 and 6.2.

Parameter	Symbol	Model Value
Link 1 length	a_1	300 mm
Link 2 length	a_2	200 mm
Outer radius ($a_1 + a_2$)	r_{max}	500 mm
Inner radius ($\lvert a_1 - a_2 \rvert$)	r_{min}	100 mm
Vertical stroke (Δd_3)	dz	150 mm
Joint 1 range	θ_1	$\pm 150°$
Joint 2 range	θ_2	$\pm 145°$
Total work volume (V)	V	$\approx 113,097$ cm^3

Table 3: Workspace parameters for the analyzed SCARA robot model.

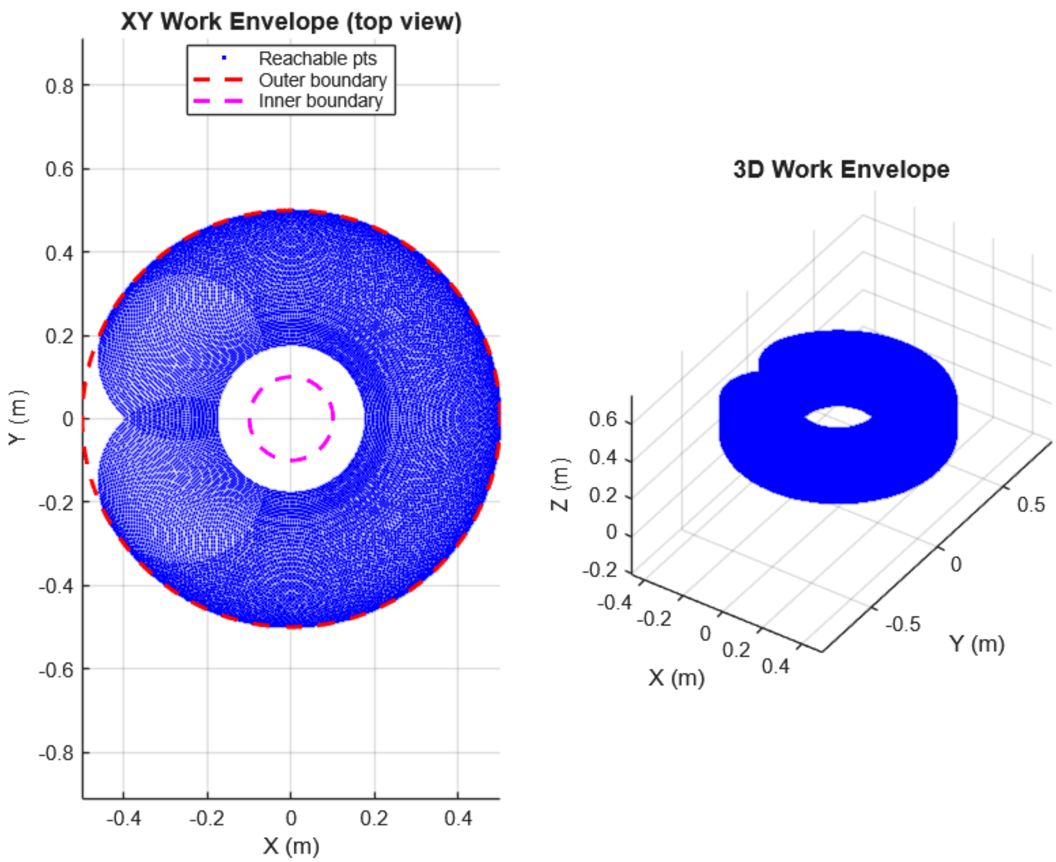

Figure 4: SCARA Work Envelope. Left: XY annular projection showing reachable points within joint limits. Right: 3D representation of the annular cylindrical workspace over the prismatic stroke.

7 Dynamics

The dynamic model describes the relationship between joint torques/forces and the resulting motion. We derive the equations of motion using the Lagrangian formulation, which is well-suited to symbolic analysis.

7.1 Lagrangian Formulation

The Lagrangian L is defined as the difference between the total kinetic energy T and the total potential energy V:

$$L = T - V \tag{7.1}$$

The equations of motion for each generalized coordinate q_i are given by the Euler-Lagrange equation:

$$\tau_i = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} \tag{7.2}$$

Collecting these for all joints yields the standard manipulator equation-of-motion form:

$$\tau = M(\mathbf{q})\ddot{\mathbf{q}} + C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) \tag{7.3}$$

where $\tau \in \mathbb{R}^4$ is the generalized force/torque vector, $M(\mathbf{q}) \in \mathbb{R}^{4\times4}$ is the inertia matrix, $C(\mathbf{q}, \dot{\mathbf{q}}) \in \mathbb{R}^{4\times4}$ is the Coriolis/centrifugal matrix, and $\mathbf{g}(\mathbf{q}) \in \mathbb{R}^4$ is the gravity vector.

7.2 Kinetic Energy and Inertia Matrix

For each link i, the kinetic energy contribution is:

$$T_i = \frac{1}{2} m_i \mathbf{v}_{ci}^T \mathbf{v}_{ci} + \frac{1}{2}\omega_i^T I_{ci} \omega_i \tag{7.4}$$

where m_i is the link mass, $\mathbf{v}_{ci}$ is the velocity of the center of mass, and I_{ci} is the inertia tensor. Neglecting vertical coupling for clarity, the dominant terms in $M(\mathbf{q})$ are:

$$M(\mathbf{q}) = \begin{bmatrix} M_{11} & M_{12} & 0 & 0 \\ M_{12} & M_{22} & 0 & 0 \\ 0 & 0 & m_3 & 0 \\ 0 & 0 & 0 & I_4 \end{bmatrix} \tag{7.5}$$

The elements are defined as:

$$\begin{aligned} M_{11} &= I_1 + I_2 + m_1 l_{c1}^2 + m_2(a_1^2 + l_{c2}^2 + 2a_1 l_{c2}\cos\theta_2) \\ M_{22} &= I_2 + m_2 l_{c2}^2 \\ M_{12} &= I_2 + m_2(l_{c2}^2 + a_1 l_{c2}\cos\theta_2) \end{aligned} \tag{7.6}$$

where l_{ci} is the distance from joint i to the center of mass of link i.

7.3 Coriolis and Centrifugal Terms

The Coriolis and centrifugal matrix is computed via Christoffel symbols of the first kind:

$$c_{ijk} = \frac{1}{2}\left(\frac{\partial M_{ij}}{\partial q_k} + \frac{\partial M_{ik}}{\partial q_j} - \frac{\partial M_{jk}}{\partial q_i}\right) \tag{7.7}$$

The (i, j) element of $\mathbf{C}$ is:

$$C_{ij} = \sum_k c_{ijk}\dot{q}_k \tag{7.8}$$

Evaluating Eq. (7.8) for the SCARA: $M(\mathbf{q})$ depends on $\mathbf{q}$ only through θ_2 (via the $\cos\theta_2$ terms in M_{11} and M_{12}), so the only non-zero partial derivatives are:

$$\begin{aligned}
\frac{\partial M_{11}}{\partial \theta_2} &= -2a_1 l_{c2} m_2 \sin\theta_2 \\
\frac{\partial M_{12}}{\partial \theta_2} &= -a_1 l_{c2} m_2 \sin\theta_2 \\
\frac{\partial M_{22}}{\partial \theta_2} &= 0 \qquad \text{(no } \theta_2 \text{ dependence)}
\end{aligned}$$

All other $\partial M_{ij}/\partial q_k = 0$. Substituting into Eq. (7.7) to obtain the non-zero Christoffel symbols:

$$\begin{aligned}
c_{112} &= \tfrac{1}{2}\left(\frac{\partial M_{11}}{\partial \theta_2} + 0 - 0\right) = -a_1 l_{c2} m_2 \sin\theta_2 \equiv -h \\
c_{121} &= c_{112} = -h \qquad \text{(by symmetry: } c_{ijk} = c_{ikj}\text{)} \\
c_{122} &= \tfrac{1}{2}\left(\frac{\partial M_{12}}{\partial \theta_2} + \frac{\partial M_{12}}{\partial \theta_2} - 0\right) = -h \\
c_{211} &= \tfrac{1}{2}\left(0 + 0 - \frac{\partial M_{11}}{\partial \theta_2}\right) = +a_1 l_{c2} m_2 \sin\theta_2 \equiv +h
\end{aligned}$$

where $h \equiv a_1 l_{c2} m_2 \sin\theta_2$. Forming $C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}}$ via Eq. (7.8) — that is, summing $c_{ijk}\dot{q}_k$ over k for each row i:

$$\begin{aligned}
\text{Row 1:} \quad & (c_{112} + c_{121})\dot{\theta}_1\dot{\theta}_2 + c_{122}\dot{\theta}_2^2 = -h\left(2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2\right) \\
\text{Row 2:} \quad & c_{211}\dot{\theta}_1^2 = +h\dot{\theta}_1^2 \\
\text{Rows 3,4:} \quad & 0 \quad \text{(joints 3 and 4 decouple from } \theta_2\text{)}
\end{aligned}$$

The Coriolis and centrifugal result, Eq. (7.9), follows directly.

Defining $h = m_2 a_1 l_{c2} \sin\theta_2$, the non-zero torques are:

$$C(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} = \begin{bmatrix} -h\dot{\theta}_2(2\dot{\theta}_1 + \dot{\theta}_2) \\ h\dot{\theta}_1^2 \\ 0 \\ 0 \end{bmatrix} \tag{7.9}$$

7.4 Gravity Vector

A key advantage of the SCARA geometry is that no gravitational torque acts on the horizontal joints (1 and 2). The prismatic joint must support the weight:

$$\mathbf{g}(\mathbf{q}) = \begin{bmatrix} 0 \\ 0 \\ -(m_3 + m_4)g_{grav} \\ 0 \end{bmatrix} \tag{7.10}$$

Sign convention note: the entry $-(m_3 + m_4)g$ in row 3 of $\mathbf{g}(\mathbf{q})$ is the force that joint 3 must exert to support the load against gravity. The negative sign is consistent with the DH convention in which d_3 increases in the $-Z_0$ (downward) direction. To hold the load stationary, the actuator must push upward (positive force), which appears as a negative gravity term in the equations of motion framework.

7.5 Complete Equation of Motion

Combining the terms, we obtain the 4-DOF SCARA dynamics:

$$\begin{aligned}
\tau_1 &= M_{11}\ddot{\theta}_1 + M_{12}\ddot{\theta}_2 - h(2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2) \\
\tau_2 &= M_{12}\ddot{\theta}_1 + M_{22}\ddot{\theta}_2 + h\dot{\theta}_1^2 \\
f_3 &= m_3\ddot{d}_3 - (m_3 + m_4)g_{grav} \\
\tau_4 &= I_4\ddot{\theta}_4
\end{aligned} \tag{7.11}$$

Remark — Friction and Damping. Equations (7.11) assume ideal joints with no friction or damping. A more complete model includes viscous damping $b_i\dot{q}_i$ and Coulomb friction $\tau_i^f = f_i^c \,\mathrm{sign}(\dot{q}_i)$ for each joint, extending Eq. (7.3) to:

$$\boldsymbol{\tau} = \mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{C}(\mathbf{q}, \dot{\mathbf{q}})\dot{\mathbf{q}} + \mathbf{g}(\mathbf{q}) + \mathbf{B}\dot{\mathbf{q}} + \boldsymbol{\tau}^f \tag{7.12}$$

where $\mathbf{B} = \mathrm{diag}(b_1, b_2, b_3, b_4)$ is the diagonal viscous damping matrix. Typical SCARA joints have viscous coefficients $b_i \approx 0.5$–5 N·m·s/rad and Coulomb friction torques of 1–10 N·m. At high operating speeds, these terms are non-negligible and must be included in the controller feedforward model.

Numerical Example. To illustrate the magnitudes, consider a SCARA with:

$$a_1 = 0.300 \text{ m}, \quad a_2 = 0.200 \text{ m}, \quad m_1 = 3.0 \text{ kg}, \quad m_2 = 2.0 \text{ kg}, \quad l_{c1} = 0.150 \text{ m}, \quad l_{c2} = 0.100 \text{ m}$$

$$I_1 = 0.050 \text{ kg}\cdot\text{m}^2, \quad I_2 = 0.020 \text{ kg}\cdot\text{m}^2, \quad I_4 = 0.005 \text{ kg}\cdot\text{m}^2$$

Evaluating at $\theta_2 = 45°$ ($\cos 45° = \sin 45° \approx 0.707$):

$$\begin{aligned}
M_{11} &= 0.050 + 0.020 + 3.0(0.150)^2 + 2.0\big(0.300^2 + 0.100^2 + 2(0.300)(0.100)(0.707)\big) \\
&= 0.1375 + 0.2848 = 0.4223 \text{ kg}\cdot\text{m}^2
\end{aligned}$$

$$M_{22} = 0.020 + 2.0(0.100)^2 = 0.0400 \text{ kg}\cdot\text{m}^2$$

$$M_{12} = 0.020 + 2.0\big(0.100^2 + (0.300)(0.100)(0.707)\big) = 0.0824 \text{ kg}\cdot\text{m}^2$$

$$h = (0.300)(0.100)(2.0)(0.707) = 0.04242 \text{ kg}\cdot\text{m}^2$$

The inertia matrix at $\theta_2 = 45°$ is therefore:

$$\mathbf{M}(\mathbf{q}) = \begin{bmatrix} 0.4223 & 0.0824 & 0 & 0 \\ 0.0824 & 0.0400 & 0 & 0 \\ 0 & 0 & 1.0 & 0 \\ 0 & 0 & 0 & 0.005 \end{bmatrix} \quad (\text{kg}\cdot\text{m}^2 \text{ or kg})$$

For joint velocities $\dot{\theta}_1 = 1.0$ rad/s and $\dot{\theta}_2 = 0.5$ rad/s, the Coriolis and centrifugal torques are:

$$\begin{aligned}
\tau_1^{\text{cor}} &= -h(2\dot{\theta}_1\dot{\theta}_2 + \dot{\theta}_2^2) \\
&= -0.04242(1.0 + 0.25) = -0.0530 \text{ N}\cdot\text{m}
\end{aligned}$$

$$\begin{aligned}
\tau_2^{\text{cor}} &= h\dot{\theta}_1^2 \\
&= 0.04242(1.0)^2 = +0.0424 \text{ N}\cdot\text{m}
\end{aligned}$$

These magnitudes (∼50 mN·m) are modest at low speeds but scale quadratically with joint velocity. At $\dot{\theta}_2 = 3$ rad/s and $\dot{\theta}_1 = 2$ rad/s, $\tau_1^{\text{cor}} \approx -0.54$ N·m — roughly 10–20% of rated joint torque, which is significant for high-speed trajectory tracking.

8 Conclusions

This report presented a complete analytical treatment of the 4-DOF SCARA robot. The key findings include:

- **Kinematics:** Closed-form expressions for forward and inverse kinematics (elbow-up/down) facilitate real-time control.
- **Singularities:** Critical operational constraints occur at $\theta_2 = 0, \pi$, which define the limits of the robot's dexterity.
- **Work Envelope:** The reachable workspace is characterized as a cylindrical annulus, with boundaries defined by the link lengths (a_1, a_2) and the prismatic stroke.
- **Dynamics:** Joints 1 and 2 are dynamically coupled, while joints 3 and 4 are decoupled. The lack of gravity torque on horizontal joints remains the SCARA's primary mechanical advantage for high-speed assembly.

A MATLAB Simulation and Visualization Scripts

The following MATLAB scripts were developed to validate the kinematic derivations and visualize the SCARA robot's performance.

A.1 Script 1: Robot Configuration Schematic

This function generates a 3D wireframe of the robot in any specified joint configuration to verify the physical link offsets and joint orientations.

```
function SCARA_schematic(th1_deg, th2_deg, d3, th4_deg)
    if nargin < 1, th1_deg=30; th2_deg=45; d3=0.05; th4_deg=0; end

    % Parameters (in meters)
    a1=0.300; a2=0.200; d1=0.400; d4=0.050;
    th1=deg2rad(th1_deg); th2=deg2rad(th2_deg); th4=deg2rad(th4_deg);

    % Joint positions (derived from FK, Equations 3.8)
    O0 = [0; 0; d1];                                   % Base (Top of Column)
    O1 = O0 + [a1*cos(th1); a1*sin(th1); 0];           % Elbow Joint
    O2 = O1 + [a2*cos(th1+th2); a2*sin(th1+th2); 0]; % Wrist (Prismatic Start)
    O3 = [O2(1); O2(2); d1 - d3 - d4];                 % Tool Tip (End-Effector)

    figure('Color','w'); hold on; grid on; axis equal;
    xlabel('X (m)'); ylabel('Y (m)'); zlabel('Z (m)');
    title('SCARA Robot Configuration','FontSize',14);
    view(30,25);

    % Link Visualization
    plot3([0 O0(1)],[0 O0(2)],[0 O0(3)],'k-','LineWidth',4);  % Base column
    plot3([O0(1) O1(1)],[O0(2) O1(2)],[O0(3) O1(3)],'b-','LineWidth',5); % Link 1
    plot3([O1(1) O2(1)],[O1(2) O2(2)],[O1(3) O2(3)],'r-','LineWidth',5); % Link 2
    plot3([O2(1) O3(1)],[O2(2) O3(2)],[O2(3) O3(3)],'g-','LineWidth',3); % Joint 3

    % Joint Markers
    pts = [0 0 0; O0'; O1'; O2'; O3'];
    scatter3(pts(:,1),pts(:,2),pts(:,3),80,'ko','filled');
    legend({'Base column','Link 1 (a1)','Link 2 (a2)','Prismatic (d3)'},'Location','best');
end
```

Listing 1: SCARA_schematic.m

A.2 Script 2: 3D Work Envelope Visualization

This script samples the reachable joint space to visualize the annular cylindrical workspace and calculate the total work volume.

```
function SCARA_workspace()
    a1=0.300; a2=0.200; d1=0.400; d4=0.050;
    d3_min=0.00; d3_max=0.150;
    th1_lim=deg2rad(150); th2_lim=deg2rad(145);

    % Sample joint space
    N=200;
    th1_v=linspace(-th1_lim, th1_lim, N);
    th2_v=linspace(-th2_lim, th2_lim, N);
```

```
    d3_v =linspace(d3_min, d3_max, 50);

    % Forward Kinematics
    [TH1,TH2]=meshgrid(th1_v,th2_v);
    PX = a1*cos(TH1) + a2*cos(TH1+TH2);
    PY = a1*sin(TH1) + a2*sin(TH1+TH2);

    figure('Color','w');
    % Subplot 1: XY Projection
    subplot(1,2,1);
    scatter(PX(:),PY(:),1,'b.'); hold on; axis equal; grid on;
    th_ring=linspace(0,2*pi,360);
    plot((a1+a2)*cos(th_ring),(a1+a2)*sin(th_ring),'r--','LineWidth',2);
    plot(abs(a1-a2)*cos(th_ring),abs(a1-a2)*sin(th_ring),'m--','LineWidth',2);
    title('XY Work Envelope (top view)');

    % Subplot 2: 3D Visualization
    subplot(1,2,2); hold on; grid on; axis equal;
    for d3=d3_v
        pz = d1 - d3 - d4;
        scatter3(PX(:),PY(:),pz*ones(numel(PX),1),1,'b.');
    end
    title('3D Work Envelope'); view(35,25);

    % Analytical Volume Calculation (Equation 6.3)
    r_max=a1+a2; r_min=abs(a1-a2); dz=d3_max-d3_min;
    V=pi*(r_max^2 - r_min^2)*dz;
    fprintf('Approx. work volume = %.4f m^3 = %.1f cm^3\n', V, V*1e6);
end
```

Listing 2: SCARA_workspace.m

A.3 Script 3: Forward Kinematics Trajectory Simulation

This function generates the end-effector path trace for a continuous joint motion profile.

```
function SCARA_kinematics_sim()
    a1=0.300; a2=0.200; d1=0.400; d4=0.050; d3=0.075;
    t=linspace(0,4*pi,400);
    th1=0.8*sin(t); th2=0.6*sin(2*t);

    % FK implementation (Eq. 3.8)
    px = a1*cos(th1) + a2*cos(th1+th2);
    py = a1*sin(th1) + a2*sin(th1+th2);
    pz = (d1 - d3 - d4)*ones(size(t));

    figure('Color','w'); hold on; grid on; axis equal;
    plot3(px,py,pz,'b-','LineWidth',2);
    scatter3(px(1),py(1),pz(1),80,'go','filled');
    scatter3(px(end),py(end),pz(end),80,'rs','filled');
    title('SCARA End-Effector Trajectory (FK simulation)');
    legend({'EE path','Start','End'},'Location','best');
    view(40,30);
end
```

Listing 3: SCARA_kinematics_sim.m